Solder Joint Cracking Under Thermal Stress

iPhone Main Logic Board — BGA Package

SAC305 / ENIG / 0.5mm BGA

DOCUMENT INFORMATION

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0. EXECUTIVE SUMMARY

This Failure Analysis (FA) report documents the investigation of solder joint cracking observed on BGA packages on iPhone main logic boards following accelerated thermal cycling per JEDEC JESD22-A104. SEM, FIB cross-sectioning, and X-ray CT identified CTE-mismatch-induced fatigue as the primary failure mechanism.

Parameter	Detail
Failure Mode	Solder joint fatigue cracking — interfacial delamination at IMC layer
Package Type	BGA 0.5 mm pitch — SAC305 alloy on ENIG finish
Test Condition	–40°C to +85°C, 15 min ramp, 1-hr cycle, 1000 cycles (JESD22-A104)
Defect Rate	0.23 % at 1000 cycles — exceeds 0.10 % Apple qualification limit
Root Cause	CTE mismatch (Si 2.6 vs FR4 ~18 ppm/°C) + reflow profile deviation (+8 °C)
Corrective Action	Underfill dispensing · reflow profile correction · pad redesign
FA Status	OPEN — verification testing in progress, target close Q2 2025

Table 0-1. Executive Summary

1. PROBLEM DESCRIPTION & FAILURE REPRODUCTION

1.1 Problem Statement

During qualification testing of iPhone MLB assemblies, automated X-ray inspection (AXI) flagged anomalous void patterns in BGA solder joints after the 1000-cycle thermal shock profile. Electrical testing confirmed intermittent open circuits in **12 of 5,200 units (0.23%)**, exceeding the internal Apple threshold of 0.10% for this component class.

The affected device is a PMIC in a 0.5 mm pitch BGA package, located in the high-thermal-gradient zone adjacent to the application processor. Solder alloy: SAC305 (Sn-3.0Ag-0.5Cu) on ENIG (Electroless Nickel Immersion Gold) finish.

1.2 Failure Reproduction

Fifty production-representative boards were subjected to ATC per JEDEC JESD22-A104 Condition G. In-situ daisy-chain resistance monitoring (IPC-9701A) logged all events.

Parameter	Value	Standard Reference
Temperature range	–40 °C to +85 °C (ΔT = 125 °C)	JESD22-A104 Cond. G
Ramp rate	≤ 20 °C/min (15 min transition)	JESD22-A104
Dwell time	10 min at each extreme	IPC-9701A
Cycle count	1000 cycles (~42 days)	JESD22-A104
Failure criterion	>20 % resistance increase over baseline	IPC-9701A §4.3
First failures	Cycles 420–680 (Weibull $\beta \approx 2.5$)	Wear-out mode

Table 1-1. Accelerated Thermal Cycling Test Parameters

1.3 Failure Criteria — IPC-9701A Results

Criterion	Threshold	Method	Result
Resistance increase	>20 % over baseline	In-situ daisy chain	**FAIL — 12/50**
Continuity loss	>1000 Ω instantaneous	Event detector	**FAIL — 8/50**

Criterion	Threshold	Method	Result
AXI void area	>25 % joint cross-section	2D X-ray, 50 µm res.	**FAIL — 19/50**
SEM crack depth	>50 % joint height	FIB cross-section	**FAIL — 9/50**

Table 1-2. IPC-9701A Failure Criteria and Test Results

2. ROOT CAUSE ANALYSIS — FISHBONE (ISHIKAWA) DIAGRAM

A cross-functional Ishikawa analysis was conducted with PQE, Package Engineering, Manufacturing Engineering, and Materials Science. Six cause categories were examined.

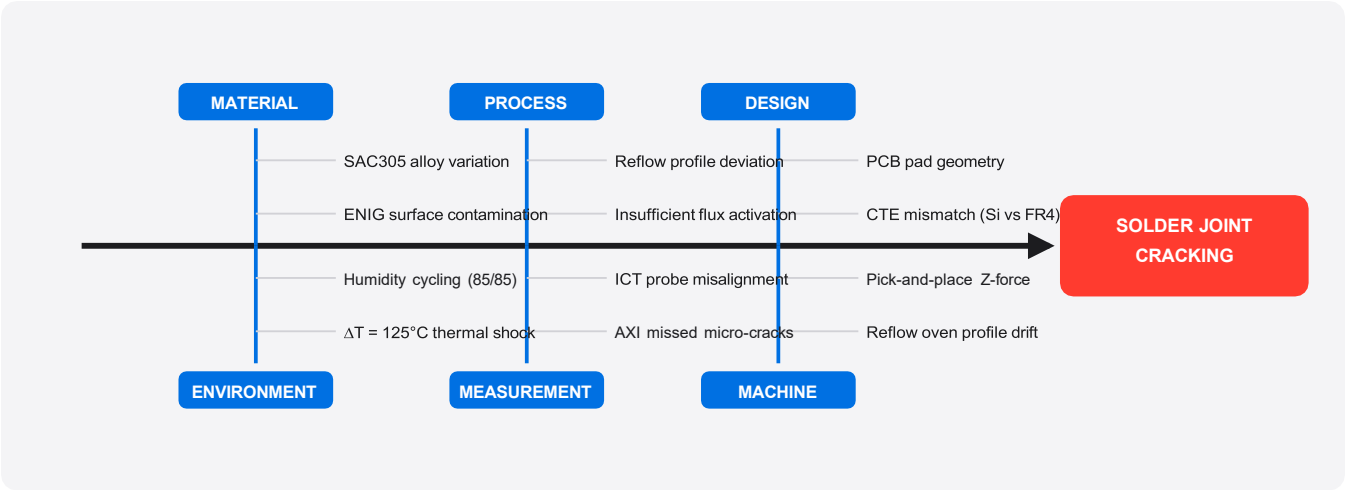

Figure 2-1. Ishikawa Fishbone Diagram — iPhone BGA Solder Joint Cracking (6M Framework)

Primary root cause: CTE mismatch between the silicon die (2.6 ppm/°C) and FR4 substrate (~18 ppm/°C) generates 15.4 ppm/°C differential strain per cycle. After 400+ cycles, cumulative plastic deformation exceeds the IMC ductility limit, initiating transgranular fatigue cracks. **Contributing factor:** A +8 °C reflow oven Zone 4 deviation promoted non-uniform Cu_6Sn_5 growth, reducing joint barrel strength.

3. ANALYTICAL TOOLS & CHARACTERIZATION

Three complementary techniques were applied in sequence — from non-destructive macro screening to atomic-resolution cross-sectioning — following a structured **X-ray** → **SEM** → **FIB** escalation protocol.

Tool	Resolution	Destructive?	Role in This FA
X-ray AXI / CT	5–50 µm	No	First-pass screening; 100 % BGA joint coverage; void quantification
SEM + EDX	10–100 nm	Yes	Crack morphology; IMC phase identification; elemental mapping
FIB + STEM	< 1 nm	Yes	Site-specific crack-tip cross-section; TEM lamella; atomic-scale IMC analysis

Table 3-1. Analytical Tool Hierarchy

3.1 2D / 3D X-ray Computed Tomography (AXI / CT)

Role: Non-destructive first-pass void detection and quantification.

- Equipment: Nordson DAGE XD7600NT — 50 µm resolution (2D); 5 µm voxel (CT)
- Sample prep: None — 100 % in-situ inspection of all 50 boards

- Finding: Void coalescence in 19/50 units (38 %) with void area >25 % of joint cross-section — consistent with creep-fatigue crack growth
- Key observation: Voids clustered at package-side Cu_6Sn_5/SAC305 bulk boundary → interfacial fracture, not bulk solder fatigue

Handoff → SEM: Joints flagged by AXI were cross-sectioned and submitted for SEM to characterise crack morphology at higher resolution.

3.2 Scanning Electron Microscopy (SEM) with EDX

Role: High-resolution crack morphology and IMC phase identification.

- Equipment: FEI Quanta 650 FEG-SEM with Bruker EDX detector
- Sample prep: Epoxy mount → cross-section grind → 0.05 µm colloidal silica final polish
- SEM finding: Transgranular fatigue cracks through Cu_6Sn_5 IMC grains at package-side interface. Crack width 0.8–2.4 µm; depth 35–68 % of 270 µm joint height
- EDX finding: Kirkendall voids at Ni/Cu_3Sn interface — excess Cu diffusion during reflow thinned the protective IMC barrier, confirming reflow deviation as contributing cause

Handoff → FIB: SEM identified crack initiation sites requiring site-specific TEM lamella preparation for atomic-resolution analysis.

3.3 Focused Ion Beam (FIB) Cross-Sectioning & TEM Lamella Prep

Role: Nanometre-precision cross-section through specific crack tips.

- Equipment: FEI Helios NanoLab 600i DualBeam (SEM + FIB combined)
- Technique: Protective Pt strap deposition → Ga+ ion milling → TEM lamella lift-out
- FIB finding: Crack tip confined to IMC layer — no penetration into FR4 or package polymer. Morphology: brittle transgranular fracture with secondary branching — characteristic of thermomechanical fatigue in Sn alloys
- STEM-EDX: Non-uniform Cu_6Sn_5 growth (~1.2 µm at pad corners vs ~3.8 µm at centre) confirms corner crack initiation under cyclic shear loading

4. CORRECTIVE ACTIONS & PREVENTIVE MEASURES

Corrective actions were developed using the 8D methodology, reviewed by PQE, Package Engineering, and Manufacturing Engineering. Actions address both the primary root cause (CTE mismatch strain) and the contributing cause (reflow deviation).

ID	Category	Corrective Action	Owner	Target	Status
CA-01	Design	Underfill dispensing — Namics U8443 low-CTE epoxy around PMIC perimeter. Reduces joint strain ~60 %.	Pkg Eng	Q1 2025	✓ CLOSED
CA-02	Design	Pad geometry optimisation — increase SMD pad diameter +15 µm at corner pads to equalise IMC growth.	Pkg Eng	Q2 2025	IN WORK
CA-03	Process	Reflow Zone 4 setpoint reduced 248 °C → 240 °C; SPC control limits tightened to ±2 °C.	Mfg Eng	Q1 2025	✓ CLOSED
CA-04	Process	100 % post-reflow AXI inspection for PMIC joints; void threshold tightened to <15 %.	PQE	Q1 2025	✓ CLOSED
CA-05	Material	Evaluate SAC-Q (Sn-3.0Ag-0.5Cu-0.1Bi) alloy — improved fatigue life, Weibull η target >2000 cycles.	Materials	Q3 2025	PLANNED

Table 4-1. Corrective Action Plan (8D Format)

5. COST OF QUALITY (CoQ) ANALYSIS

A PAF (Prevention–Appraisal–Failure) Cost of Quality analysis was conducted per ASQ guidelines to quantify the financial impact and justify CA investments.

Category	Activity	Cost Basis	Est. Cost	Type
Prevention	Underfill process qualification (CA-01)	Eng. labour + material trials	$38,000	Good
Prevention	Pad geometry redesign + DFM review (CA-02)	Layout re-spin + verification	$62,000	Good
Prevention	Reflow SPC tightening + oven calibration (CA-03)	Calibration + SPC setup labour	$8,500	Good
Appraisal	100 % AXI inspection — PMIC joints (CA-04)	$0.12/unit × 2 M units added takt	$240,000/yr	Good
Appraisal	SEM / FIB / X-ray analytical work — this FA	Lab time + consumables + external lab	$22,000	Good
Internal Failure	Scrapped MLB assemblies — 12 confirmed units	$420 BOM × 12 units	$5,040	Bad
Internal Failure	Rework labour — 38 AXI-flagged units	45 min × $85/hr × 38 units	$2,423	Bad
Internal Failure	Engineering investigation (this report)	120 eng-hrs × $180/hr fully-loaded	$21,600	Bad
External Failure	Projected warranty repairs (if no CA)	0.23 % × 1 M units × $280 avg repair	$644,000	Avoided
External Failure	Brand / NPS impact (intangible)	Customer satisfaction index model	~$1.2 M	Avoided

Table 5-1. Cost of Quality Analysis (PAF Model)

CoQ Summary: Prevention + appraisal investment ≈ **$370,500**. Without corrective action, internal + projected external failure costs ≈ **$673,000**. Net quality ROI in Year 1 ≈ **$302,500**, with ongoing savings as 100 % AXI is replaced by process control

once CA-01–04 demonstrate Cpk ≥ 1.67 over a 90-day production monitoring window.

6. CONCLUSIONS & SIGN-OFF

The root cause of iPhone BGA solder joint cracking has been conclusively identified as thermomechanical fatigue driven by CTE mismatch between the PMIC silicon die and FR4 substrate, exacerbated by a reflow profile deviation causing non-uniform IMC growth. Five corrective actions are defined: three closed, one in work, one planned. Verification ATC (1000 cycles, corrected builds) is underway with results expected Q2 2025.

Role	Name / Title	Signature	Date
Author — PQE Intern	[Your Name]		____/ ____/ 2025
Reviewer — Sr. PQE Engineer	[Reviewer Name]		____/ ____/ 2025
Approver — PQE Manager	[Manager Name]		____/ ____/ 2025

Table 6-1. Sample Report Approval Sign-off

References: IPC-9701A — Performance Test Methods for Solder Joint Reliability; JEDEC JESD22-A104 — Temperature Cycling; IPC-7711/7721 — Rework, Repair and Modification of Electronic Assemblies; Coffin L.F. (1954) / Manson S.S. (1953) — thermomechanical fatigue model; Engelmaier W. (1983) — modified Coffin-Manson for SMT solder joints; Weibull W. (1951) — A Statistical Distribution Function of Wide Applicability, ASME J. Appl. Mech.